# Ninth International Olympiad in Linguistics

## Pittsburgh (United States of America), 24–31 July 2011

### Team Contest Problem

The following ten lines are incorrect examples of Sanskrit poetry. They were originally written correctly, but there have been five deleted macra, four added macra, three changed letters and two deleted words, so that only one line remained unchanged. No syllables have been added or lost (except in the deleted words).

For instance, sentence 9 was originally ***syād indravajrā yadi tau jagau gaḥ***. We can restore the deleted macron over the ***a*** in ***syād*** for metrical reasons, but to purge the added macron over the second ***a*** in ***indravajrā*** requires knowing the word (or comparison with line 10). Fortunately, the poets wrote in such a way that it's possible to correct all the changes except that one without any prior knowledge of Sanskrit.

The mark "¯", called a macron (pl. macra), denotes vowel length; ***bʰ***, ***dʰ***, ***gʰ***, ***ḥ***, ***j***, ***ñ***, ***ṇ***, ***ś***, ***tʰ*** and ***y*** are consonants. There is one more rule of transliteration relevant to the metre that you will have to discover.

The translations correspond to the lines after the two words were deleted and the three letters changed, but before any macra were added or deleted.

| | | |
|---|---|---|
| 1. | ***bʰujanga-prayātam caturbʰir gakaraiḥ*** | "The movement of the snake" consists of four *ga*s. |
| 2. | ***gurunidʰanamānulagʰur iha śāśikalā*** | In the case where a *guru* is at the end of 14 *lagʰu*s, it is said to be "the ascending period of the moon". |
| 3. | ***jarau jarau tato jagau ca pañcacamaram vadet*** | A *ja*-and-*ra*, a *ja*-and-*ra* and then a *ja*-and-*ga* is called "the fan made of five yak tails". |
| 4. | ***mabʰalagā gajagatiḥ*** | "The gait of an elephant" is *ma bʰa la ga*. |
| 5. | ***mo go go go vidyunmālā*** | A *ma* and a *ga* and a *ga* and a *ga* is "the garland of lightning". |
| 6. | ***nanagi madʰumati*** | Where there is *na na ga*, there is "(that which is) full of honey". |
| 7. | ***pramaṇikā*** _____ _____ | "The little measure" is _____ _____. |
| 8. | ***pramāṇikā padadvayam vadanti pañcacāmaram*** | Two lines of "the little measure" they call "the fan made of five yak tails". |
| 9. | ***syad indravājrā yadi tau jagau gaḥ*** | If perchance there are a pair of *ta*s, a *ja*-and-*ga*, and a *ga*, then it is "Indra's thunderbolt". |
| 10. | ***ūpendravajrā pratʰame lagʰau sā*** | "Upendra's thunderbolt" is that (Indra's thunderbolt) in which the first is *lagʰu*. |

**(a)** What is the additional rule of transliteration?

**(b)** Restore the two deleted words, revert the three changed letters, remove the four added macra, and restore the five deleted macra.

**(c)** The macra have been removed from the Sanskrit mnemonic ***yamatarajabʰanasalagam***. Which syllables were *guru*?

⚠ A mnemonic is a word or sentence that helps remembering something (*How I wish I could recollect* … → 3.14159 ≈ $\pi$). Indra and his younger brother Upendra are Hindu gods. —*Adam Hesterberg*

## Distribute at 0 minutes

The following insights will at some point be given as hints: the meaning of *guru*, the constraints of the poetry, and (in 3 hints) the use of ***yamatarajab$^h$anasalagam***.

## Answers as of 30 minutes

Team: _____

**(a)** What is the additional rule of transliteration?

_____.

**(b)**
- Deleted words: _____, _____.
- Changed letters:
  1.
  2.
  3.
- Words with added macra:
  1. The second ***a*** in the word ***indravājrā*** in line 9.
  2.
  3.
  4.
- Words with deleted macra:
  1. The ***a*** in the word ***syad*** in line 9.
  2.
  3.
  4.
  5.

**(c)** Which syllables in ***yamatarajab$^h$anasalagam*** were *guru*?

**(+)** Insights into the problem matter (use the back of the paper if necessary):